



CODECHEF

The CodeChef.com Challenge—your monthly dose of coding puzzles from India's biggest coding contest, now in print!

Welcome back to the CodeChef Puzzle Challenge, sponsored by LINUX For You and Directi. It's time to take a look at another one of CodeChef's interesting problems for March, and while you're at it, let's see if you can win some cash as well!

Last month's puzzle

The Fibonacci number series is a famous one where $F[n] = F[n-1] + F[n-2]$. The first few numbers of this series are 0, 1, 1, 2, 3, 5, 8, and so on. Considering 1-based indices, what is the 999th Fibonacci number? (For example, The 7th Fibonacci number is 8).

The solution

Considering 1 based indices the 999th Fibonacci number is:

```
16602747662452097049541800472897
70183494805119838482806235855309
1918573717701170201065510185595
8986051040947369188792784622330
159810295229978363112326187605
391990367653997999267314332397
18860373345088375054249
```

How did we do it?

This number can be calculated

by using big-integer libraries provided by some languages like Java, or by using a language like Python that supports large calculations natively. You can also write code for this in C/C++ by using arrays to store the numbers.

And the winners for February are...

- Srinath Krishnan
- Balakrishnan Erode
- Aaliya.Ash

Puzzle for March

Four distinct balls can be partitioned into three non-empty subsets in an unordered manner as follows.

```
{(1), (2), (3,4)}
{(1,2), (3), (4)}
{(1), (2,3), (4)}
{(1,4), (2), (3)}
{(1,3), (2), (4)}
{(2,4), (1), (3)}
```

We see that there are six ways of partitioning the four balls. In how many ways can we partition five such balls into three non-empty subsets?

Would you like to get a little extra cash?

Then solve this month's CodeChef Challenge puzzle and mail your answer to codechef@efyindia.com by March 18, 2010. Three lucky winners will win cash of Rs 1,000 each! 

About CodeChef: CodeChef.com is India's first, non-commercial, online programming competition, featuring monthly contests in more than 35 different programming languages. CodeChef has been created by Directi, as a way to continuously challenge and engage the developer community. The goal is to provide a platform for practice, competition and improvement as well as enable developers to benchmark their skills against their peers. Log on to www.codechef.com for the next monthly challenge that takes place from March 1-11, 2010, and win cash prizes worth Rs.55,000.